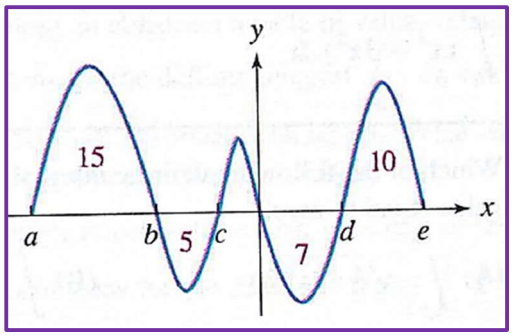
Worksheet 6-9 Test Prep FTC and Integrals
AP Calculus Name_____

Questions with an asterisk (*) indicate that a calculator may be used to solve the problem. Otherwise, a calculator is not permitted.

The figure to the right shows the graph of f. The areas of the regions between the graph of f and the x-axis are labeled and it is known that $\int_c^0 \boldsymbol{f(x)dx = 3}$. Determine the value of each definite integral.



1.) $\int_0^e f(x)\,dx$

2.) $\int_b^d f(x)\,dx$

3.) $\int_a^0 |f(x)|\,dx$

4.) $\left|\int_a^0 f(x)\,dx\right|$

5.) The function f is continuous and $\int_4^{19} f(u)du = 10$. What is the value of $\int_1^4 [x \cdot f(x^2+3)]\,dx$?

[A] $\frac{5}{2}$ [B] 5 [C] 10 [D] 20 [E] 40

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*6.) If $F'(x) = \sqrt{1+x^3}$ and $F(1) = 5$, then $F(3) =$

[A] 1.230 [B] 3.585 [C] 6.230 [D] 8.535 [E] 11.230

7.) If $F(x) = \int_2^x \sqrt{t^3-1}\,dt$, then $F'(2) =$

[A] 0 [B] $\frac{1}{2\sqrt{7}}$ [C] $\frac{6}{\sqrt{7}}$ [D] $\sqrt{7}$ [E] $12\sqrt{7}$

x	1	2	3	5	6
$F(x)$	4	12	2	8	20
$F'(x)$	–1	3	7	2	15

*8.) Use the table above to compute $\int_1^3 F'(x)\,dx$

[A] –2 [B] 15 [C] 8 [D] 6 [E] 2

*9.) Use the table above to compute $\int_1^3 F'(2x)\,dx$

[A] 8 [B] 2 [C] –2 [D] 4 [E] 16